MSG No. 131.440 $P4'_2/m'm'c$ [Type III, tetragonal]

Table 1: Wyckoff site: `2a`, site symmetry: `m'm'm'`.

No.	position	mapping
1	$[0, 0, 0]$	`[1,2,3,4,13,14,15,16]`
2	$\left[0, 0, \frac{1}{2}\right]$	`[5,6,7,8,9,10,11,12]`

Table 2: Wyckoff site: `2b`, site symmetry: `m'm'm'`.

No.	position	mapping
1	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[1,2,3,4,13,14,15,16]`
2	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[5,6,7,8,9,10,11,12]`

Table 3: Wyckoff site: `2c`, site symmetry: `m'm'm'`.

No.	position	mapping
1	$\left[0, \frac{1}{2}, 0\right]$	`[1,2,3,4,13,14,15,16]`
2	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[5,6,7,8,9,10,11,12]`

Table 4: Wyckoff site: `2d`, site symmetry: `m'm'm'`.

No.	position	mapping
1	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[1,2,3,4,13,14,15,16]`
2	$\left[\frac{1}{2}, 0, 0\right]$	`[5,6,7,8,9,10,11,12]`

Table 5: Wyckoff site: `2e`, site symmetry: `-4m'2'`

No.	position	mapping
1	$\left[0, 0, \frac{1}{4}\right]$	`[1,4,5,6,11,12,14,15]`
2	$\left[0, 0, \frac{3}{4}\right]$	`[2,3,7,8,9,10,13,16]`

Table 6: Wyckoff site: `2f`, site symmetry: `-4m'2'`

No.	position	mapping
1	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{4}\right]$	`[1,4,5,6,11,12,14,15]`
2	$\left[\frac{1}{2}, \frac{1}{2}, \frac{3}{4}\right]$	`[2,3,7,8,9,10,13,16]`

Table 7: Wyckoff site: `4g`, site symmetry: `2m'm'`.

No.	position	mapping
1	$[0, 0, z]$	`[1,4,14,15]`
2	$[0, 0, -z]$	`[2,3,13,16]`

continued ...

Table 7

No.	position	mapping
3	$\left[0,\, 0,\, \frac{1}{2}-z\right]$	`[5,6,11,12]`
4	$\left[0,\, 0,\, z+\frac{1}{2}\right]$	`[7,8,9,10]`

Table 8: Wyckoff site: `4h`, site symmetry: `2m'm'`.

No.	position	mapping
1	$\left[\frac{1}{2},\, \frac{1}{2},\, z\right]$	`[1,4,14,15]`
2	$\left[\frac{1}{2},\, \frac{1}{2},\, -z\right]$	`[2,3,13,16]`
3	$\left[\frac{1}{2},\, \frac{1}{2},\, \frac{1}{2}-z\right]$	`[5,6,11,12]`
4	$\left[\frac{1}{2},\, \frac{1}{2},\, z+\frac{1}{2}\right]$	`[7,8,9,10]`

Table 9: Wyckoff site: `4i`, site symmetry: `2m'm'`.

No.	position	mapping
1	$\left[0,\, \frac{1}{2},\, z\right]$	`[1,4,14,15]`
2	$\left[0,\, \frac{1}{2},\, -z\right]$	`[2,3,13,16]`
3	$\left[\frac{1}{2},\, 0,\, \frac{1}{2}-z\right]$	`[5,6,11,12]`
4	$\left[\frac{1}{2},\, 0,\, z+\frac{1}{2}\right]$	`[7,8,9,10]`

Table 10: Wyckoff site: `4j`, site symmetry: `m'2m'`.

No.	position	mapping
1	$[x,\, 0,\, 0]$	`[1,2,15,16]`
2	$[-x,\, 0,\, 0]$	`[3,4,13,14]`
3	$\left[0,\, -x,\, \frac{1}{2}\right]$	`[5,7,10,12]`
4	$\left[0,\, x,\, \frac{1}{2}\right]$	`[6,8,9,11]`

Table 11: Wyckoff site: `4k`, site symmetry: `m'2m'`.

No.	position	mapping
1	$\left[x,\, \frac{1}{2},\, \frac{1}{2}\right]$	`[1,2,15,16]`
2	$\left[-x,\, \frac{1}{2},\, \frac{1}{2}\right]$	`[3,4,13,14]`
3	$\left[\frac{1}{2},\, -x,\, 0\right]$	`[5,7,10,12]`
4	$\left[\frac{1}{2},\, x,\, 0\right]$	`[6,8,9,11]`

Table 12: Wyckoff site: `4l`, site symmetry: `m'2m'`.

No.	position	mapping
1	$\left[x,\, 0,\, \frac{1}{2}\right]$	`[1,2,15,16]`
2	$\left[-x,\, 0,\, \frac{1}{2}\right]$	`[3,4,13,14]`
3	$[0,\, -x,\, 0]$	`[5,7,10,12]`

continued ...

Table 12

No.	position	mapping
4	$[0,\, x,\, 0]$	`[6,8,9,11]`

Table 13: Wyckoff site: `4m`, site symmetry: `m'2m'`.

No.	position	mapping
1	$\left[x,\, \frac{1}{2},\, 0\right]$	`[1,2,15,16]`
2	$\left[-x,\, \frac{1}{2},\, 0\right]$	`[3,4,13,14]`
3	$\left[\frac{1}{2},\, -x,\, \frac{1}{2}\right]$	`[5,7,10,12]`
4	$\left[\frac{1}{2},\, x,\, \frac{1}{2}\right]$	`[6,8,9,11]`

Table 14: Wyckoff site: `8n`, site symmetry: `..2'`

No.	position	mapping
1	$\left[x,\, x,\, \frac{1}{4}\right]$	`[1,11]`
2	$\left[x,\, -x,\, \frac{3}{4}\right]$	`[2,10]`
3	$\left[-x,\, x,\, \frac{3}{4}\right]$	`[3,9]`
4	$\left[-x,\, -x,\, \frac{1}{4}\right]$	`[4,12]`
5	$\left[x,\, -x,\, \frac{1}{4}\right]$	`[5,15]`
6	$\left[-x,\, x,\, \frac{1}{4}\right]$	`[6,14]`
7	$\left[-x,\, -x,\, \frac{3}{4}\right]$	`[7,13]`
8	$\left[x,\, x,\, \frac{3}{4}\right]$	`[8,16]`

Table 15: Wyckoff site: `8o`, site symmetry: `.m'.`

No.	position	mapping
1	$[0,\, y,\, z]$	`[1,14]`
2	$[0,\, -y,\, -z]$	`[2,13]`
3	$[0,\, y,\, -z]$	`[3,16]`
4	$[0,\, -y,\, z]$	`[4,15]`
5	$\left[y,\, 0,\, \frac{1}{2} - z\right]$	`[5,11]`
6	$\left[-y,\, 0,\, \frac{1}{2} - z\right]$	`[6,12]`
7	$\left[-y,\, 0,\, z + \frac{1}{2}\right]$	`[7,9]`
8	$\left[y,\, 0,\, z + \frac{1}{2}\right]$	`[8,10]`

Table 16: Wyckoff site: `8p`, site symmetry: `.m'.`

No.	position	mapping
1	$\left[\frac{1}{2},\, y,\, z\right]$	`[1,14]`
2	$\left[\frac{1}{2},\, -y,\, -z\right]$	`[2,13]`
3	$\left[\frac{1}{2},\, y,\, -z\right]$	`[3,16]`
4	$\left[\frac{1}{2},\, -y,\, z\right]$	`[4,15]`
5	$\left[y,\, \frac{1}{2},\, \frac{1}{2} - z\right]$	`[5,11]`

continued ...

Table 16

No.	position	mapping
6	$\left[-y,\ \frac{1}{2},\ \frac{1}{2}-z\right]$	[6,12]
7	$\left[-y,\ \frac{1}{2},\ z+\frac{1}{2}\right]$	[7,9]
8	$\left[y,\ \frac{1}{2},\ z+\frac{1}{2}\right]$	[8,10]

Table 17: Wyckoff site: 8q, site symmetry: m'..

No.	position	mapping
1	$[x,\ y,\ 0]$	[1,16]
2	$[x,\ -y,\ 0]$	[2,15]
3	$[-x,\ y,\ 0]$	[3,14]
4	$[-x,\ -y,\ 0]$	[4,13]
5	$\left[y,\ -x,\ \frac{1}{2}\right]$	[5,10]
6	$\left[-y,\ x,\ \frac{1}{2}\right]$	[6,9]
7	$\left[-y,\ -x,\ \frac{1}{2}\right]$	[7,12]
8	$\left[y,\ x,\ \frac{1}{2}\right]$	[8,11]

Table 18: Wyckoff site: 16r, site symmetry: 1

No.	position	mapping
1	$[x,\ y,\ z]$	[1]
2	$[x,\ -y,\ -z]$	[2]
3	$[-x,\ y,\ -z]$	[3]
4	$[-x,\ -y,\ z]$	[4]
5	$\left[y,\ -x,\ \frac{1}{2}-z\right]$	[5]
6	$\left[-y,\ x,\ \frac{1}{2}-z\right]$	[6]
7	$\left[-y,\ -x,\ z+\frac{1}{2}\right]$	[7]
8	$\left[y,\ x,\ z+\frac{1}{2}\right]$	[8]
9	$\left[-y,\ x,\ z+\frac{1}{2}\right]$	[9]
10	$\left[y,\ -x,\ z+\frac{1}{2}\right]$	[10]
11	$\left[y,\ x,\ \frac{1}{2}-z\right]$	[11]
12	$\left[-y,\ -x,\ \frac{1}{2}-z\right]$	[12]
13	$[-x,\ -y,\ -z]$	[13]
14	$[-x,\ y,\ z]$	[14]
15	$[x,\ -y,\ z]$	[15]
16	$[x,\ y,\ -z]$	[16]